

## Education

Vellore Institute of Technology - Vellore, India  
B.Tech – Electronics and Communication  
(2021-2025)  
• CGPA: 8.31

Delhi Public School Raj Nagar, Ghaziabad  
• Class XII: 84.4% (2018-2020)  
• Class X: 84.8% (2017-2018)

## Work Experience

- **Embedded Engineer II**

MLWorkx Pvt Ltd – Noida, India (*Mar, 2026 – Present*)

1. Conducted **post-silicon bring-ups** and system-level debugging of multiple PCBs, developed test firmware and validated GPIOs, I2C, SPI, I2S and other peripherals using **Segger J-Link, oscilloscopes, and timing analysis**.
2. **Designed, built, and shipped a complete RTC-based Digital Clock product on 8051 (ABOV A96G166FR)** using **bare-metal C**, successfully delivered to, and **50000+** units ordered by the customer.

- **Embedded Engineer I**

MLWorkx Pvt Ltd – Noida, India (*Jul, 2025 – Mar, 2026*)

1. **Designed and built an end-to-end nRF54L15 BLE-based product from scratch**, where 15 BLE devices monitor insect frequencies on individual trees, perform edge ML inference, predict insect's presence. Sends the suspected audio to a Gateway which sends it to the cloud to prevent large-scale insect infestations in high-value crops.
2. **Implemented a wireless charging solution using TI BQ51051 and a BLE range-extension solution** using the **nRF21540 FEM** for a US-based Frisbee product
3. **Shipped an ESP32-based handheld device** that triggers audio recording on NFC scan and stores data locally on an SD card, delivered as a customer-ready product.

- **Software & Electronics System (SES) Intern**

Whirlpool – Pune, India

*Feb, 2025 – Jul, 2025*

1. Automated hardware validation of the Whirlpool's Fridge Panel PCB through C# scripts by sending and receiving APIs, opcodes, and payloads to/from the MCU, reducing manual testing time from 30 min to ~3 min.

## Skills

- **Core Technical Skills:** C | Low-Level Programming | Device Drivers | RTOS | nRF SDK Connect | Zephyr RTOS | Computer Architecture (ARM Cortex-M3) | Basic Data Structures & Algorithms | BLE – Range & Throughput Optimisation | Communication Protocols: UART, I2C, I2S, SPI | Python | MQTT | Git | Linux Shell Scripting | Bash | Soldering | PCB Debugging | Nordic SoCs | ESP32 | ATmega328P | ABOV SoCs
- **Software & Tools:** Git | Keil | Abov OCD Debugger SW | VS Code | nRF Connect SDK | flux.ai PCB Designer | Github | Arduino IDE | Automation Scripting

## Projects

- **RAID-HV: Road Anomaly and Incoming Disturbance Alert System for High-Speed Vehicles:**
  - Engineered a "Vehicle to Vehicle" safety network using GPS for location tracking, an accelerometer for detecting sudden jerks, and a LoRa module with an SMA 433MHz antenna.
  - The system detects road disturbances and shares coordinates with vehicles within a 10-15 km radius.
- **Auto-Login Attendance Bot:** Programmed a smart attendance bot that autonomously attends online lectures. Crafted using Python and Bash scripting. Internship offered by Atlan's Co-founder, Prukalpa.
- **Door-side weather forecasting alert system:** Completed a NodeMCU-based alarm that alerts users when leaving the house during potential harsh weather conditions, utilizing the OpenWeatherMap API with 80-90% accuracy.

## Recognitions

- Google Code-In 2018 Finalist: Collaborated with a community of 300+ members and completed 44 tasks for SCoReLab. Featured on the Contributors page of SCoReLab.
- Top 10 Ideas - CharmhealthCodeRx Hackathon 2023: Prepared an intelligent summary system for doctors, processing over 3000 lines of Patient data in the form of XML using the OpenAI API (Python) and XML Parser.